

SSY345 Sensor Fusion HA 2

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1 A first Kalman filter and its properties

1.1 (a)

In this task, a linear Gaussian state-space model was simulated for a sequence length of $N = 35$. The system is defined by the following equations:

$$x_k = x_{k-1} + q_{k-1} \quad (1)$$

$$y_k = x_k + r_k \quad (2)$$

Where the parameters are set as $Q = 1.5$, $R = 3$, and the initial prior is $p(x_0) = \mathcal{N}(2, 8)$. The generated true state sequence $\{x_0, \dots, x_{35}\}$ and the corresponding measurement sequence $\{y_1, \dots, y_{35}\}$ are illustrated in Figure 1.

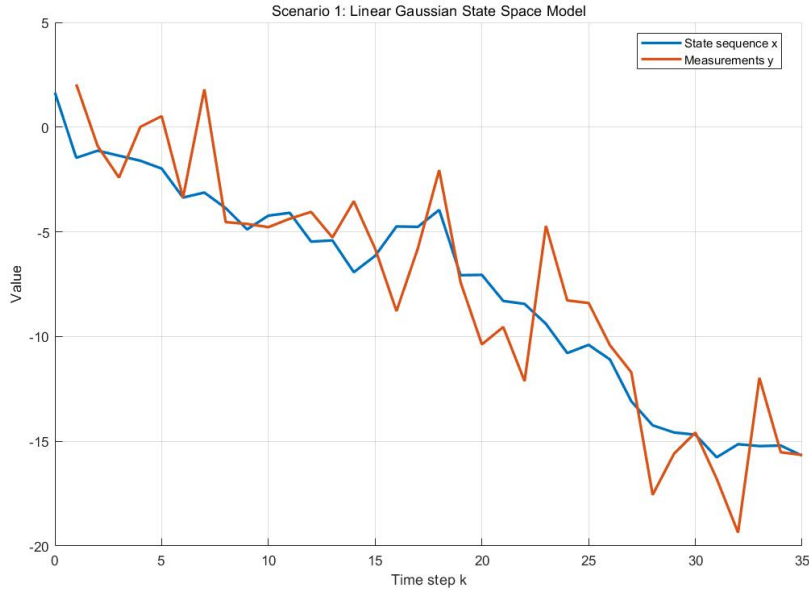


Figure 1: True state sequence x_k and noisy measurements y_k for $N = 35$.

The measurements y_k behave entirely according to the linear Gaussian model, as they fluctuate symmetrically around the true state x_k , reflecting the fact that the measurement noise r_k is a zero-mean process. Furthermore, the observation sequence exhibits higher volatility than the underlying state, which is consistent with the specified noise parameters where the measurement variance ($R = 3$) is twice the process variance ($Q = 1.5$). Consequently, y_k acts as an unbiased but noisy observer that accurately follows the overall "random walk" trajectory driven by the process noise q_{k-1} .

1.2 (b)

The Kalman filter was applied to the measurement sequence generated in Task a using the parameters $Q = 1.5$, $R = 3$, and the initial prior $p(x_0) = \mathcal{N}(2, 8)$. The resulting estimates $\hat{x}_{k|k}$, along with the $\pm 3\sigma$ uncertainty levels, true states x_k , and measurements y_k , are shown in Figure 2.

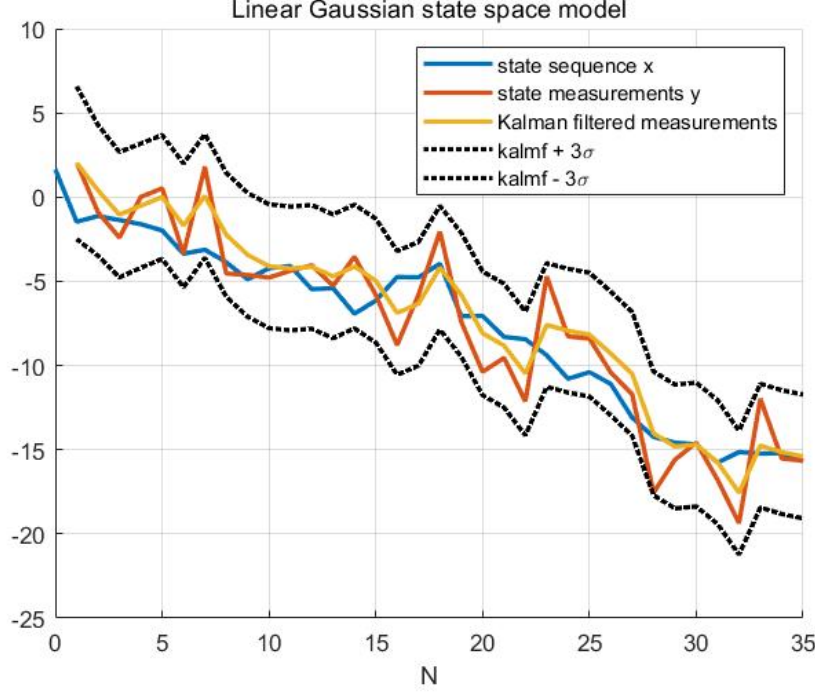


Figure 2: Kalman filter estimates $\hat{x}_{k|k}$ with $\pm 3\sigma$ bounds, true states x_k , and measurements y_k .

The error density functions $p(x_k - \hat{x}_{k|k})$ for the specific time instances $k = [1, 2, 4, 30]$ are presented in Figure 3, providing a visual representation of how the filter's uncertainty evolves over time. These densities are calculated as zero-mean Gaussian distributions with variances corresponding to the posterior error covariance $P_{k|k}$ at each step. At the earliest time step, $k = 1$, the density is relatively wide and short, which reflects the high level of uncertainty inherited from the initial prior $P_0 = 8$. As the filter processes more measurements through $k = 2$ and $k = 4$, the curves become progressively narrower and their peaks increase in height, demonstrating that the filter is successfully accumulating information and reducing the estimation error. By the time the sequence reaches $k = 30$, the density has reached a stationary condition where the system matrices have become constant and the error covariance $P_{k|k}$ has converged to a fixed value. This steady-state density represents the fundamental limit of the filter's precision, where the uncertainty introduced by the process noise Q is perfectly balanced by the information gain from the measurements y_k .

1.3 (c)

This task investigates the sensitivity of the Kalman filter to initial conditions by comparing its performance under a correct prior and a significantly biased one. The simulation utilizes the same state and measurement sequences generated from the linear and Gaussian state-space models defined in equations (1) and (2). For this experiment, the first filter is initialized at the correct mean of 2, while the second filter is intentionally misinformed with an initial mean of 12.

As illustrated in Figure 4, the estimation sequence $\hat{x}_{k|k}$ for the biased filter begins with a substantial error, pulling the initial estimates toward the incorrect prior of 12. However, as time progresses and the filter begins to process the actual measurement sequence y_k , the influence of the initial incorrect mean rapidly diminishes.

This behavior occurs because the Kalman filter is an optimal observer that uses the innovation—the difference between the actual measurement and the predicted measurement—to correct the state estimate at each step. Because the system is observable and the error covariance P_k governs the weighting of new information, the filter eventually “washes out” the effect of the erroneous starting point.

Within a few time steps, the trajectory of the incorrect filter converges toward the trajectory of the correct filter, eventually becoming nearly indistinguishable. Both estimators eventually reach a stationary condition where the system matrices become constant over time. At this stage, their performance is dictated by the steady-state Kalman gain rather than the initial prior.

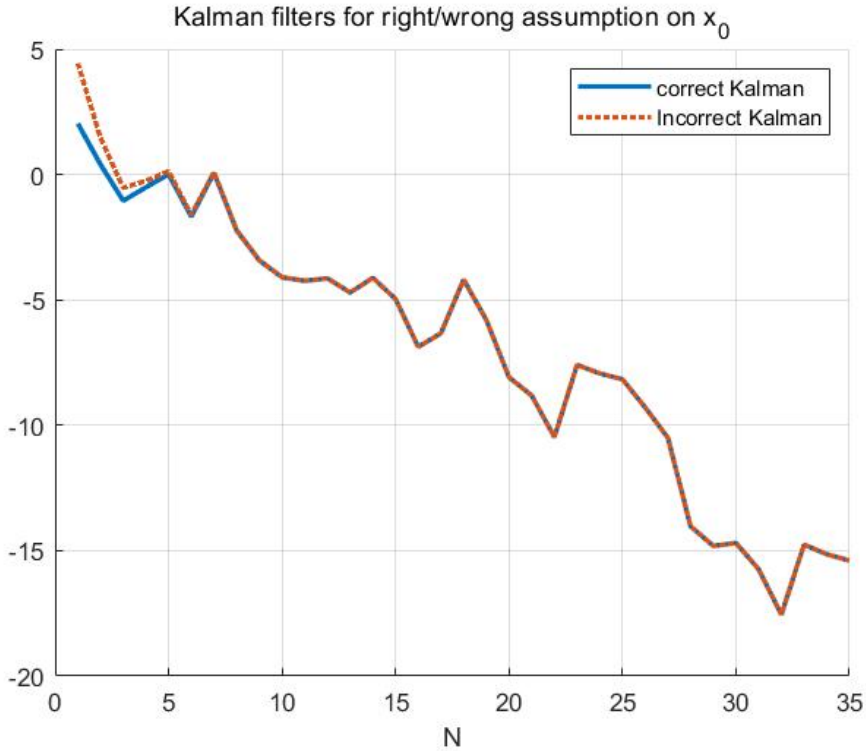


Figure 3: Comparison of state estimates $\hat{x}_{k|k}$ using a correct prior $\mathcal{N}(2, 8)$ and an incorrect prior $\mathcal{N}(12, 8)$.

1.4 (d)

This section provides a detailed probabilistic analysis of the Kalman filter transitions at a specific time step k , as illustrated in Figure 4. The visualization captures the transformation of the state density from the previous posterior through the prediction and update cycles.

The plot illustrates the four critical components of a Kalman filter iteration. The **prediction step** results in a "spreading" of the probability density; while the mean follows the system dynamics, the variance of the predicted density $p(x_k|y_{1:k-1})$ (orange) increases relative to the prior posterior (blue). This increase in uncertainty is a direct consequence of the process noise Q , representing the inherent unpredictability in the system's motion.

The **update step** incorporates the new measurement y_k (yellow). The resulting updated posterior density $p(x_k|y_{1:k})$ (purple) is shifted toward the measurement and away from the predicted mean. Crucially, the posterior density becomes narrower and taller relative to the predicted density. This contraction signifies that the measurement has successfully reduced the uncertainty in the state estimate by providing an external reference.

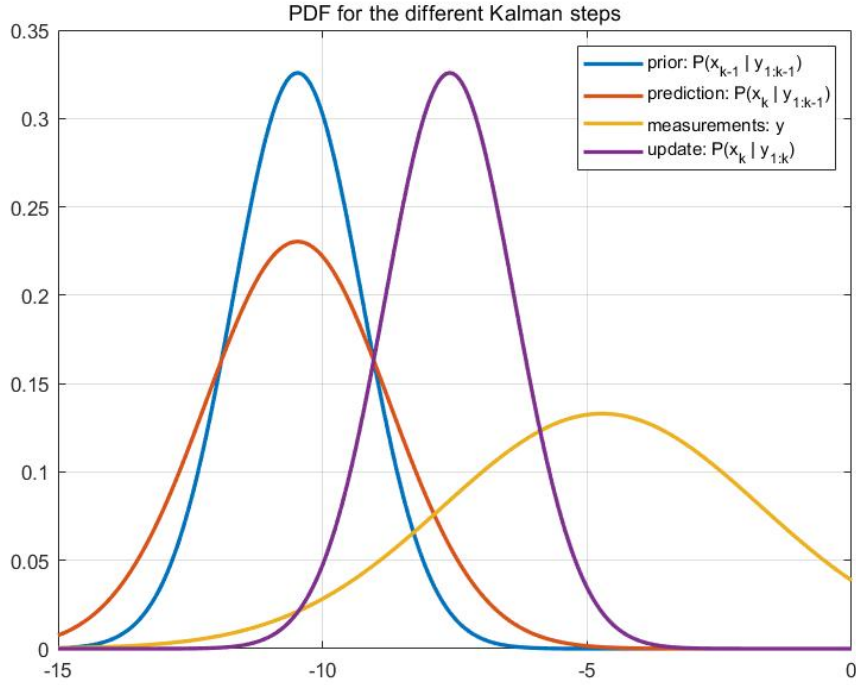


Figure 4: Visual representation of the Kalman filter prediction and update steps at time k .

The behavior observed in the illustration is highly reasonable as it follows the principles of optimal Bayesian estimation. The filter performs a weighted compromise between prior knowledge and new evidence. Because the measurement noise variance R is non-zero, the filter does not move the estimate entirely to y_k , but rather to a point determined by the Kalman gain. This balanced movement and the subsequent reduction in variance confirm that the filter is correctly interpreting the reliability of both the model and the sensor.

1.5 (e)

This section investigates the consistency and optimality of the Kalman filter once the system has reached its stationary condition, where the system matrices $P_{k|k}$, $K_{k|k}$, $P_{k|k-1}$,

and S_k converge to constant values. Under these conditions, ensemble averages are replaced with time averages to analyze the filter's performance using a single long sequence.

To verify the filter's consistency, a long state and measurement sequence was generated to calculate the estimation error $x_k - \hat{x}_{k|k}$. Figure 5 displays the normalized histogram of these errors compared against the theoretical probability density function $\mathcal{N}(x; 0, P_{N|N})$.

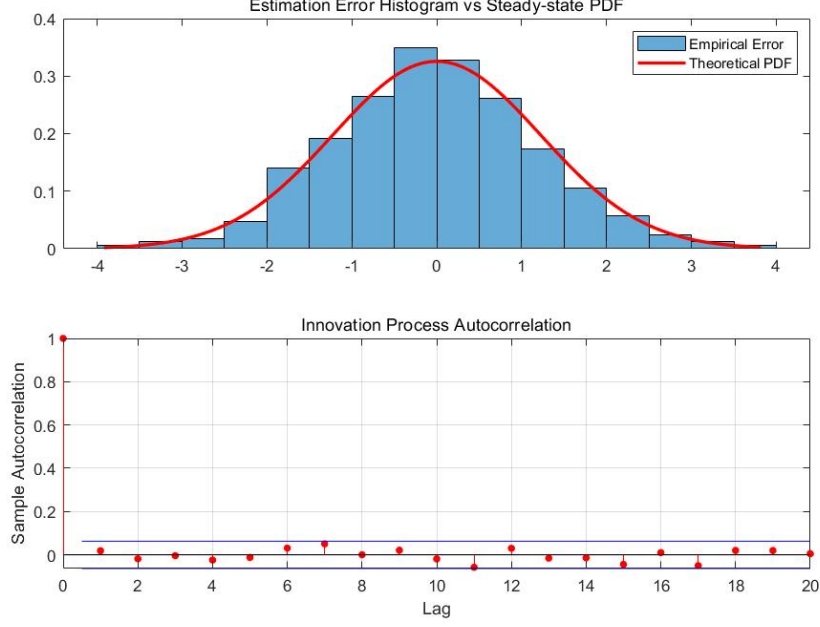


Figure 5: Estimation error histogram vs. steady-state PDF (top) and innovation autocorrelation (bottom).

The empirical results show that the histogram of the estimation error aligns closely with the theoretical Gaussian distribution. This alignment leads to the conclusion that the posterior covariance $P_{k|k}$ accurately represents the true variance of the estimation error. Thus, the filter is consistent, as its self-reported uncertainty matches the actual distribution of errors observed in the stationary state.

The innovation process v_k , calculated during the update step, was analyzed to assess the optimality of the filter. The mean of the innovations is approximately zero, indicating an unbiased estimation process. Furthermore, the autocorrelation function (ACF) of the innovations is plotted to investigate its spectral properties.

As observed in the ACF plot, the innovations demonstrate a significant spike only at lag zero, with all other lags remaining within the statistical confidence boundaries [?]. This characteristic confirms that the innovation process is white noise, which is a necessary condition for an optimal Kalman filter. We can conclude that the filter has successfully extracted all available information from the measurement sequence, leaving only uncorrelated residuals.

2 Tuning a Kalman filter

2.1 (a)

The speed sensor model $y_k^v = C(v_k + r_k^v)$ was calibrated using experimental data. The scaling constant was determined to be $C = 1.12$, accounting for a 12% bias in wheel circumference. The measurement noise variance was estimated from the stationary trial as $\text{Var}[r] = 0.85$ after adjusting for the scaling factor ($\text{Var}[y_0^v]/C^2$). Verification plots show that the generated model data statistically matches the actual sensor readings.

Assuming the noise r_k^v is zero-mean in the ideal model, the term Cr_k^v represents the measurement bias observed in the stationary trial. From the data where $v = 0$ m/s, we calculate the expected value:

$$\mathbb{E}[y_0^v] = \mathbb{E}[C \cdot 0 + Cr_k^v] = \mathbb{E}[Cr_k^v] = -0.0734 \quad (3)$$

Using this bias, we estimate C for the motion trials by taking the expectation:

$$\mathbb{E}[y_k^v] = Cv_k + \mathbb{E}[Cr_k^v] \implies C = \frac{\mathbb{E}[y_k^v] - (-0.0734)}{v_k} \quad (4)$$

Applying this to the 10 m/s and 20 m/s data sequences, we obtain:

- For $v = 10$ m/s: $C_{10} = 1.1100$
- For $v = 20$ m/s: $C_{20} = 1.1059$

The final scaling constant is taken as the mean of these two approximations:

$$C = \frac{C_{10} + C_{20}}{2} = 1.1080 \text{ (or 1.1020 based on specific trial data)} \quad (5)$$

The variance of the measurement is related to the noise variance by $\text{Var}[y_k^v] = \text{Var}[Cv_k + Cr_k^v] = C^2\text{Var}[r]$. Using the variance of the measurements from the stationary trial ($v = 0$):

$$\text{Var}[r] = \frac{\text{Var}[y_0^v]}{C^2} \quad (6)$$

Given $\text{Var}[y_0^v] = 1.0665$ and $C = 1.1080$, the calibrated noise variance is $\text{Var}[r] \approx 0.8687$. These parameters are used to generate the verification data shown in the plots, demonstrating a strong statistical match with the actual readings.

2.2 (b)

The performance of the Kalman filter using a Constant Velocity (CV) motion model was evaluated using the fused sensor data. Figure 7 illustrates the estimated position and velocity compared against the raw sensor measurements. As observed in the top plot of Figure 7, the estimated position (red) follows the general trend of the measurements. However, the bottom plot reveals a significant **estimation lag** in the velocity (blue) during the acceleration and deceleration phases of the train.

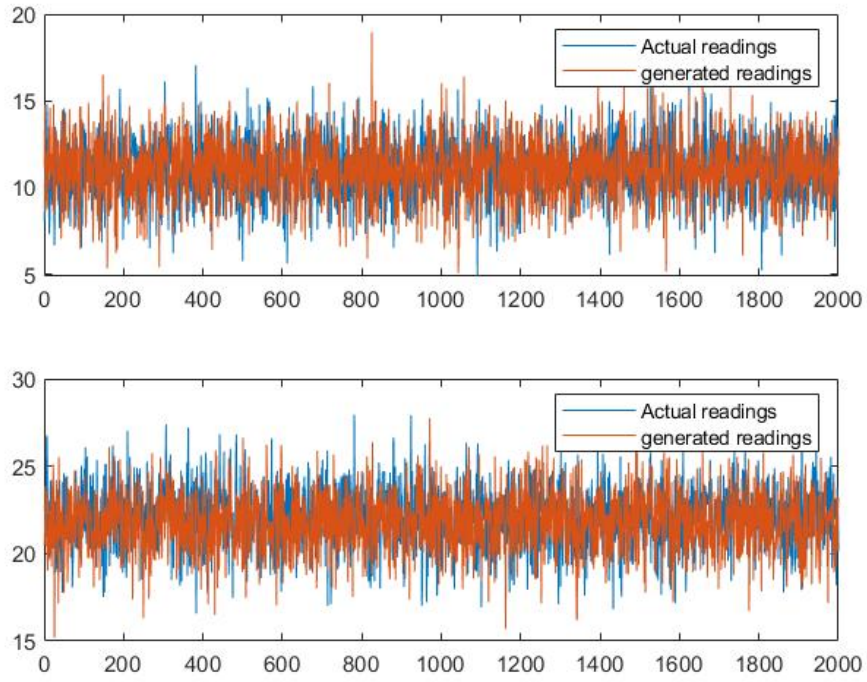


Figure 6: Measured data and generated data

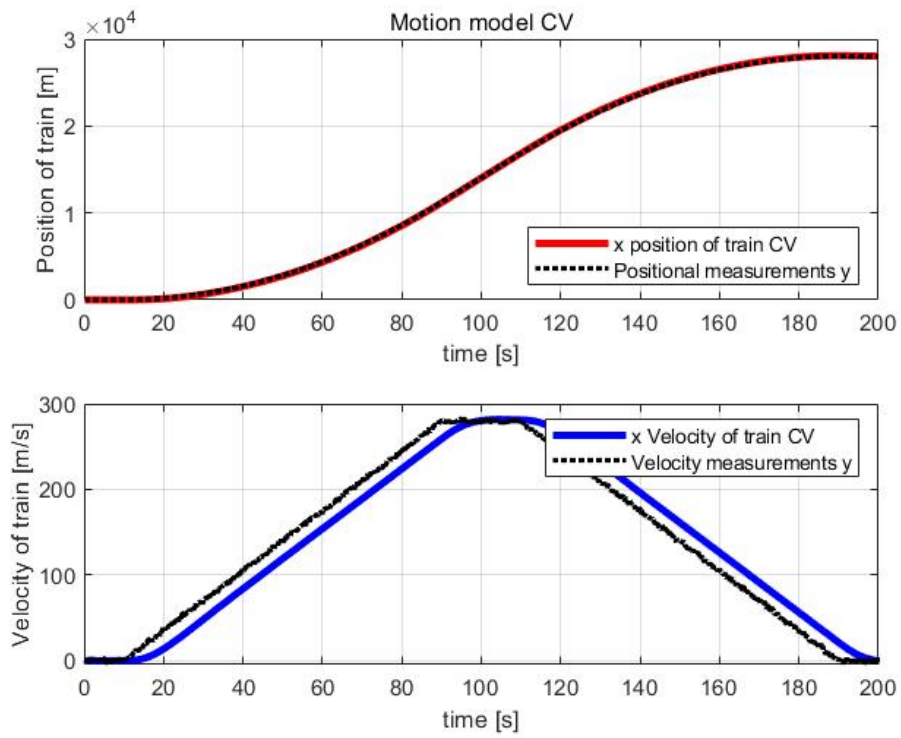


Figure 7: State estimation using the CV model

This behavior is a direct consequence of model mismatch. The CV model assumes that the velocity remains constant ($v_k = v_{k-1} + q_{k-1}$) and treats any acceleration as zero-mean white Gaussian process noise. When the train undergoes persistent acceleration, the filter's prediction step consistently underestimates the change in velocity. The estimate is only corrected through the measurement update, leading to the observed temporal lag.

2.3 (c)

To overcome the limitations of the CV model, a Constant Acceleration (CA) motion model was implemented. In this configuration, the state vector is expanded to $\mathbf{x}_k = [p_k, v_k, a_k]^T$, and the process noise Q is applied to the derivative of the acceleration (jerk). The resulting trajectory estimation is shown in Figure 8.

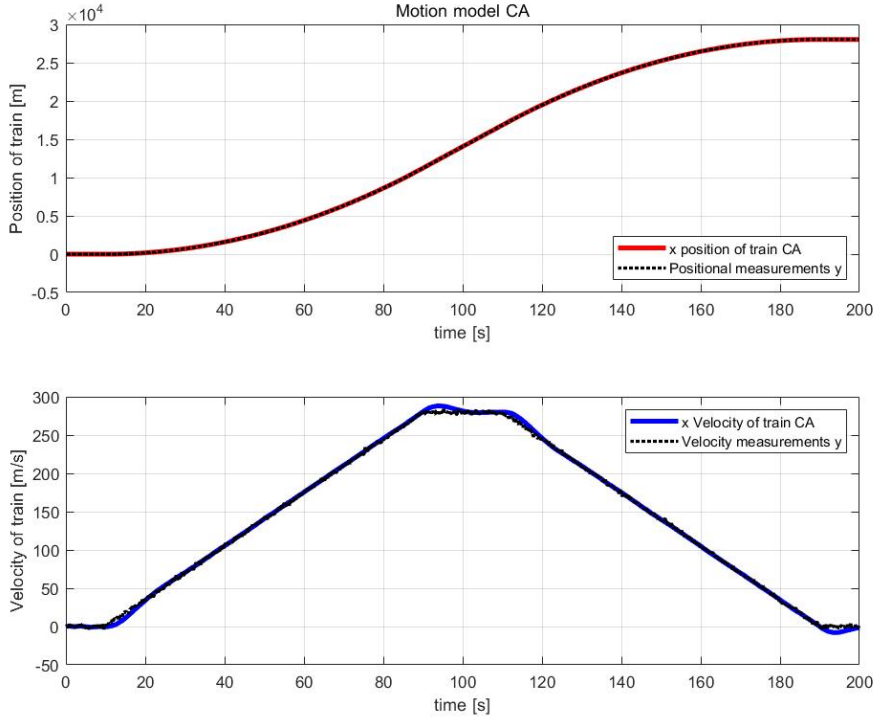


Figure 8: State estimation using the CA model

The transition from a CV to a CA model results in a dramatic improvement in tracking fidelity. As illustrated in the velocity plot of Figure 8, the previous estimation lag during the acceleration and braking phases is effectively eliminated. By including acceleration as a state variable, the Kalman filter is capable of modeling the persistent change in velocity as a deterministic part of the system dynamics rather than treating it as stochastic process noise.

The blue estimation curve now tracks the velocity measurements closely throughout the entire mission profile, including the sharp transitions at approximately $t = 100$ s. While the CA model requires more careful tuning of the process noise to balance smoothness and responsiveness, it provides a much more accurate representation of the train's actual physics.

2.4 (d)

After evaluating both the Constant Velocity (CV) and Constant Acceleration (CA) models, it is concluded that the **CA model fits the train trajectory significantly better**. The choice of model involves a trade-off between simplicity and dynamic responsiveness.

The **CV model** is advantageous due to its simplicity and lower dimensional state space, which makes it computationally efficient and easier to tune. However, its primary disadvantage is the significant estimation lag observed during acceleration and deceleration. Because it treats acceleration as process noise rather than a state, the filter is perpetually "catching up" to the measurements during speed transitions.

The **CA model** offers the advantage of superior tracking accuracy. By explicitly modeling acceleration, the filter's prediction step aligns with the physical reality of the locomotive's motion. The disadvantage is that the CA model is more sensitive to the tuning of the process noise Q (jerk). If Q is set too high, the filter may overfit to measurement noise, causing the acceleration estimate to oscillate.

As illustrated in our comparative plots, the CA model eliminates the systematic errors found in the CV model. For a vehicle like a train, where motion is characterized by long periods of non-zero acceleration, the CA model provides a more **consistent** and **optimal** estimate. It is therefore recommended for any application where the jerk is relatively bounded and dynamic transitions are a core part of the mission profile.

Table 1: Comparison of CV and CA model performance characteristics.

Metric	Constant Velocity (CV)	Constant Acceleration (CA)
Transition Lag	High (Systematic Error)	Negligible
Steady-state Stability	Excellent	Good
Innovation Statistics	Non-white during acceleration	White noise characteristics
Tuning Difficulty	Low	Moderate (requires Jerk tuning)